

OTS-02 Summary Document

Overview

This document summarises the OTS-02 lecture on what volatility, implied and realized volatility are and how to trade the volatility premium.

The following topics are covered:

- What is volatility?
- Empirical properties of volatility
- Two types of volatility
- Volatility premium
- Why does premium exist?

What is volatility?

It's well-known that predicting returns is almost impractical and impossible, but predicting volatility is certainly possible with historical data.

- A time series has (at least) two components.
- A trend (or drift): In 2016, SPY rose 11%.
- But there was considerable variability around this return.
- Informally, volatility is a way to quantify these fluctuations.

VIX (Volatility Index)

VIX is the market's expectation of volatility. That includes commodities, forex, and the stock market. Increased volatility of the stock market is usually a sign that there is a lot of uncertainty. It is generally calculated on the baseline index of the exchange like NIFTY(NSE), S&P 500(NYSE).

The following snapshot shows the CBOE VIX plot from Sep 2016 to Jan 2021



Empirical properties of volatility

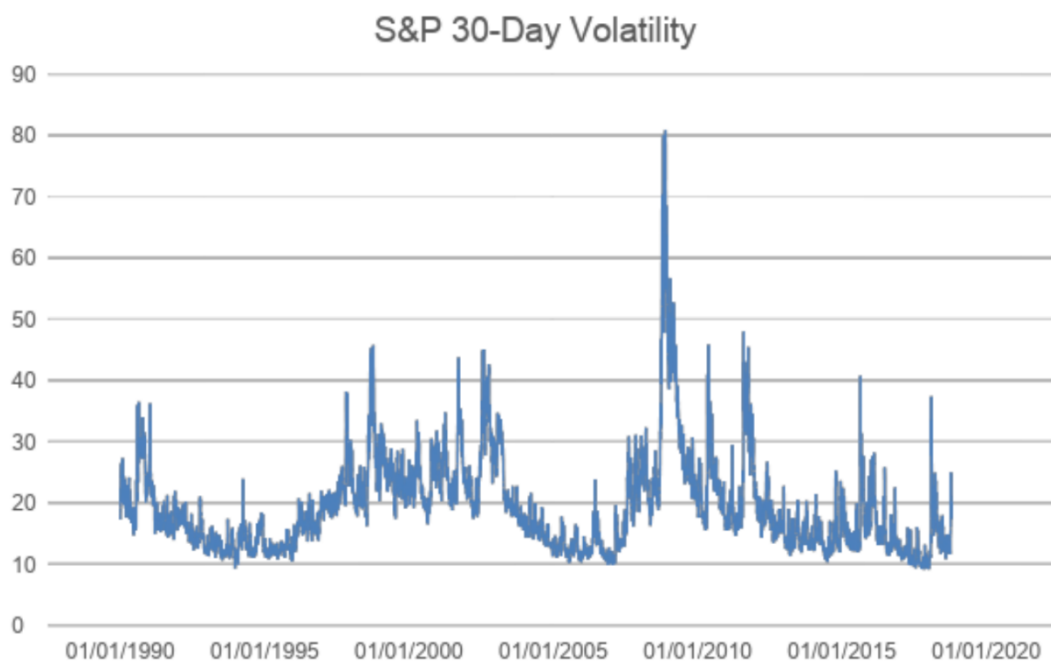
The following properties of volatility have been observed -

1. It is mean-reverting in nature.
2. Volatility is negatively correlated to returns.
3. It is generally much larger than the returns.

1. Volatility is mean-reverting

We have observed that the volatility tends to revert to the mean.

The following snapshot shows the 30-day volatility of the S&P 500.

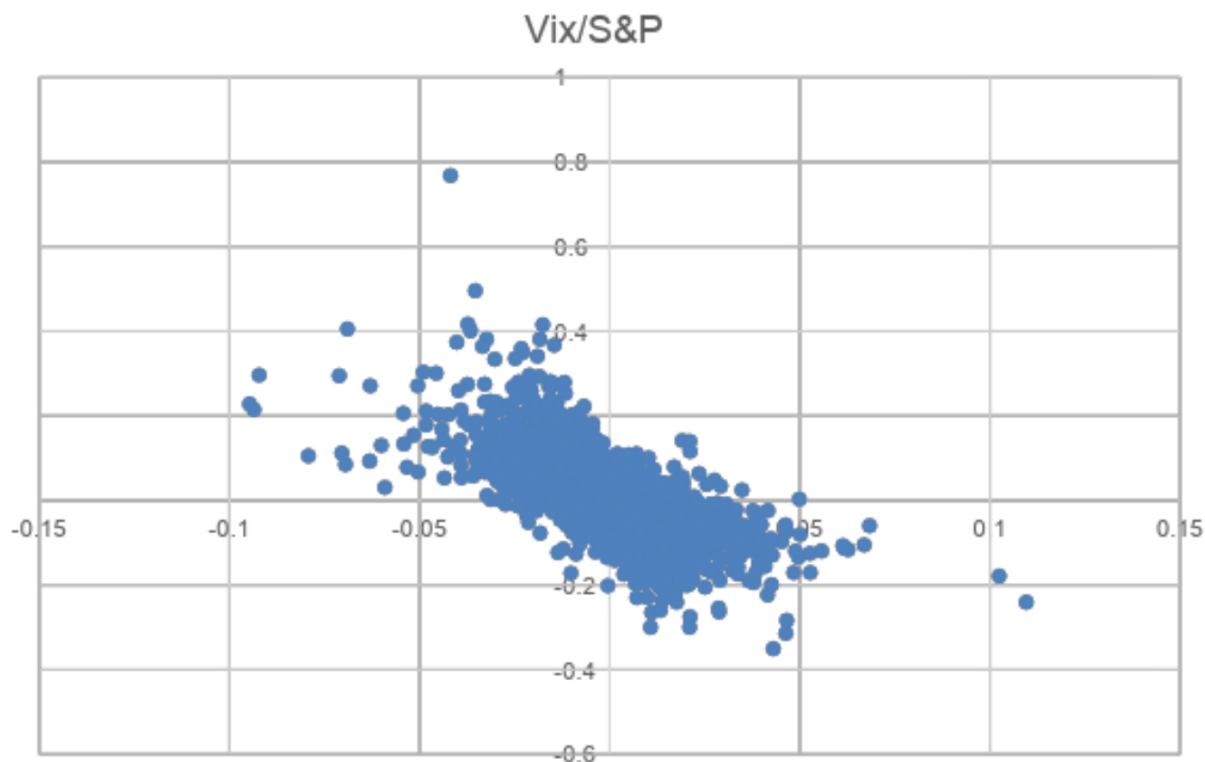


As we see, the volatility returns to mean always. This plot confirms that volatility is mean-reverting.

2. Volatility is negatively correlated with returns.

It has been observed that volatility tends to move in the opposite direction as compared to returns.

S&P 500 volatility/return correlation is -.70 (weekly is -0.71, monthly is -0.68)



Looking at the above correlation graphs, the observation is that the correlation was always negative and between -0.68 to -0.71.

This effect is less prevalent for commodities and individual stocks. Over the last five years the correlations have been.

Product	Correlation
Oil	-0.47
Gold	0.15
AAPL	-0.59
AMZN	-0.50
IBM	-0.05

Two types of volatility

There are various types of volatility.

Realised volatility is the variation of a variable measured over a period of time. When we take a historical period, it's called **historical volatility**.

The **implied volatility (IV)** is a measurement that considers the market expectations about the volatility. For a liquid options contract, it is possible to calculate the implied volatility. One can get a realistic estimation of the market expectations of the variation within the underlying asset's price in the future.

Black–Scholes is one of the foremost popular models to calculate implied volatility. In simple words, historical volatility is backward-looking whereas implied volatility is forward-looking.

Implied Volatility (IV) can be represented in 2 ways -

- IV as a function of the strike is called a “smile” or “skew.”
- IV as a function of time is called “Term structure.”

Reasons for volatility smile

- Correlation between underlying and volatility: as the stock drops, volatility increases, so lower strikes have higher implied volatilities.
- Leverage effect: as the stock price drops, a company becomes more leveraged as its debt/equity ratio increases.
- Supply and demand: people buy specific options for protection.
- Longshot bias: far OTM options are cheap in dollar terms, so they are bought as lottery tickets. This pushed the volatility up.

Reasons for the term structure

- Mean reversion: If the market is currently volatile, short-dated options will have high implied vols, but the back months will increase far less as we expect the effect to be temporary.
- Different things happen in different periods:
 - Earnings announcements
 - Takeover speculation
 - Different crops
 - Different weather patterns

Volatility premium

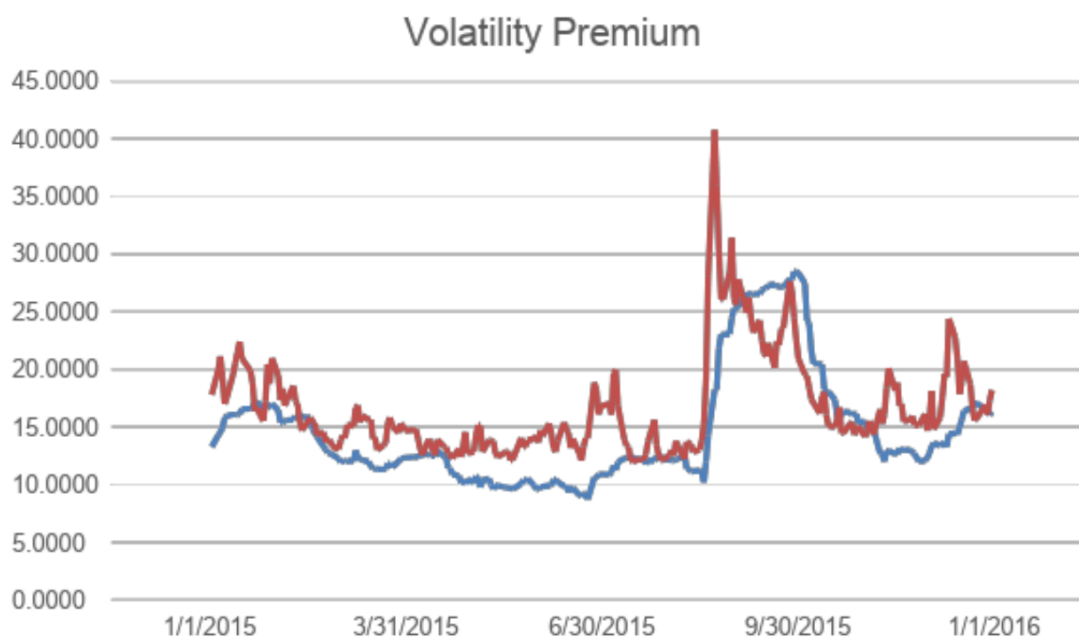
If realised volatility is not equal to implied volatility, we can trade the option and replicate it in the underlying, thus, getting a profit. So, we need to find situations where we expect realised volatility to be different from the implied volatility.

Model-driven methods are based on time series analysis, using the empirical regularities documented earlier. Situational methods can be based on unusual events such as elections or earnings, or unusual times such as weekends.

But we must also acknowledge the variance (or volatility) premium.

Usually: Implied volatility > subsequent realized volatility.

The S&P 500 implied volatility (red) and realized volatility (blue) is shown below:

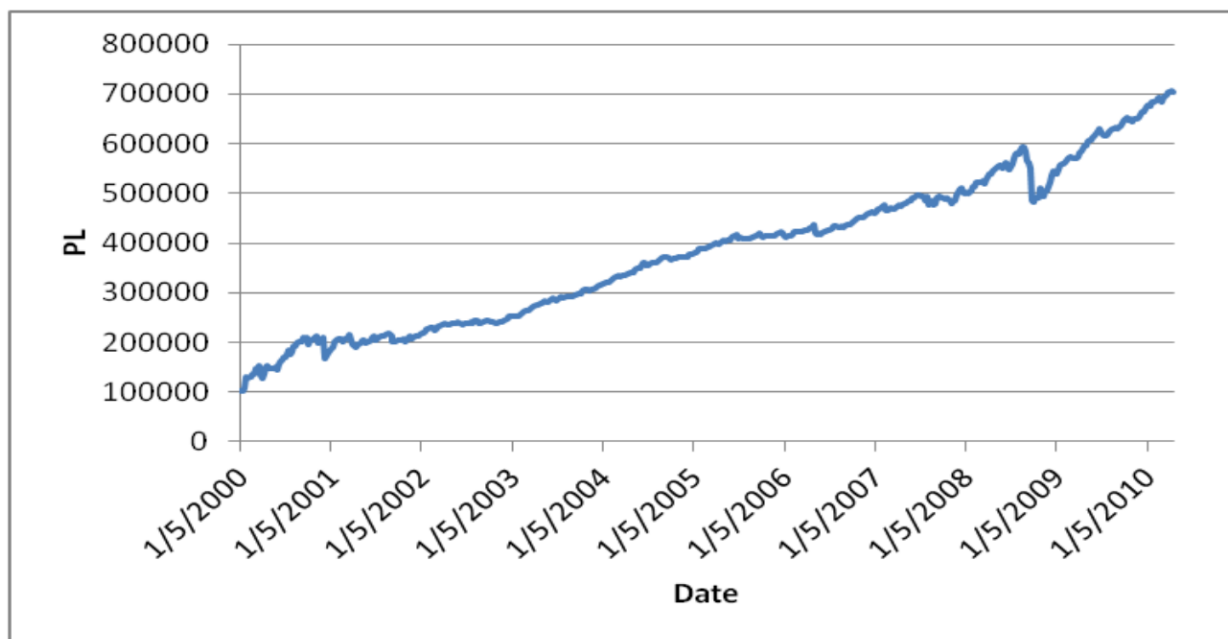


It is persistent even at very low volatility levels. As a percentage, the premium is generally higher at low volatility levels.

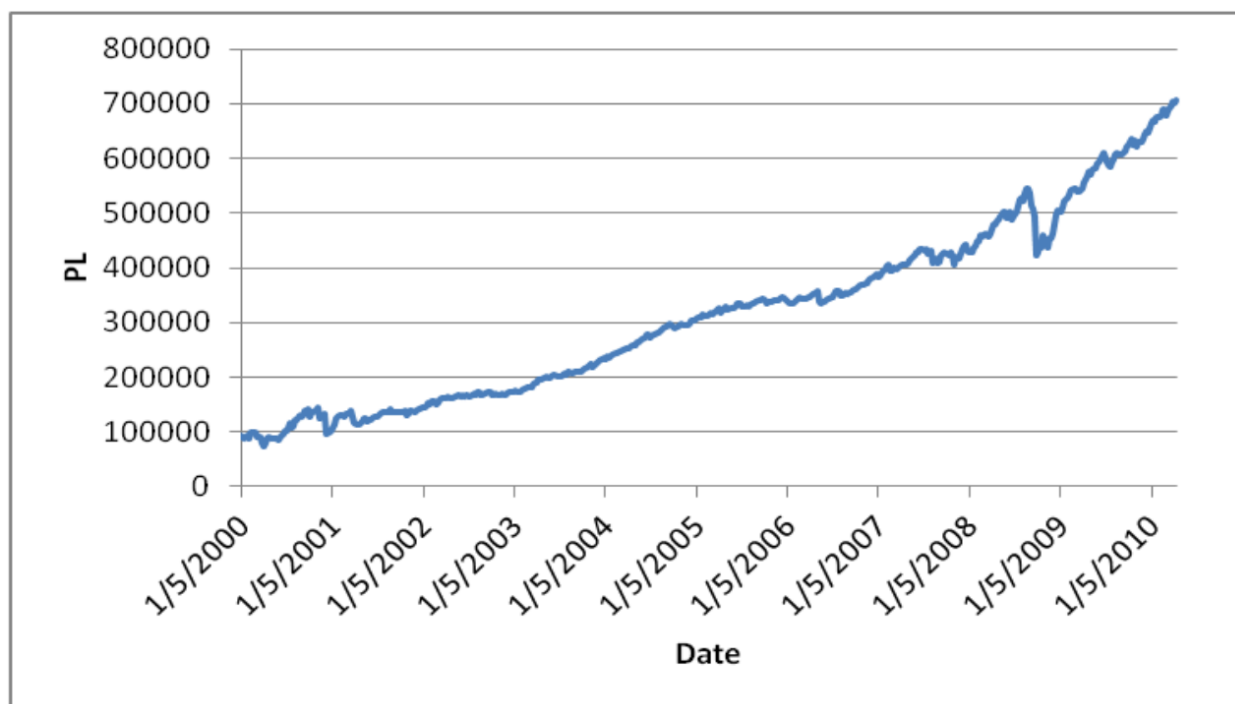
VIX Level	Volatility Premium (as %)
<15	0.27
15<20	0.26
20<25	0.20
25<30	0.20
30<35	0.18
35<40	0.14
>45	0.10

Examples

Selling 10 delta QQQ, 2nd month strangles (using account value for margin)



Selling ATM QQQ, 2nd month straddles (Same vega exposure as with strangles)



Why does premium exist?

- Both calls and puts provide insurance.
- Puts against a crash hurting an existing portfolio.
- Calls against FOMO (Fear Of Missing Out).
- All insurers charge a premium for their products so they can make a profit.
- This is no different from any other shop.
- People vastly overestimate the danger of extreme events.
- Terrorism kills 180 Americans a year (about 8 if we exclude 9/11).
- Heart disease kills 600,000 Americans each year.
- This mistake means options are overpriced.
- "Black Swans" doesn't happen very often.